



Exploratory Data Analysis: the diamonds dataset

Exam Proctor faculty pilot · Monday 5 October 2026 · Part B, 11:38 to 12:15 · Old IQAC · 40 marks

Instructions

- Work in **Kaggle**. Open the notebook address shown on the projector and click **Copy & Edit**. This makes your own copy with the diamonds data already attached.
- If Copy & Edit does not work: on kaggle.com click **Create** → **New Notebook**, then **Add Input** in the right-hand panel, search for **diamonds** (by Shivam Agrawal) and click **+**.
- You may also use the documentation at pandas.pydata.org, seaborn.pydata.org and matplotlib.org. Stay in your own notebook: other people's Kaggle notebooks and discussions, every other website, and every other app on the laptop are reported.
- Run the first cell (below) with **Shift + Enter**; the first run takes about 30 seconds while Kaggle starts. Then answer each question in its own cell, adding a text cell where a question asks you to explain.
- This is a pilot of the monitoring tool. Nobody is graded; the marks show how much detail is expected.

First cell: load the data (already in the notebook)

```
import glob
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# The diamonds dataset attached to this notebook (Kaggle dataset:
shivam2503/diamonds)
path = glob.glob('/kaggle/input/**/diamonds.csv', recursive=True)[0]
df = pd.read_csv(path, index_col=0) # the first column is only a row number

# Same category order as seaborn's built-in copy: best grade first
df['cut'] = pd.Categorical(df['cut'], ['Ideal', 'Premium', 'Very Good', 'Good',
'Fair'])
df['color'] = pd.Categorical(df['color'], ['D', 'E', 'F', 'G', 'H', 'I', 'J'])
df['clarity'] = pd.Categorical(df['clarity'], ['IF', 'VVS1', 'VVS2', 'VS1', 'VS2',
'SI1', 'SI2', 'I1'])
df.head()
```

Questions

Question 1

2 marks

After loading the data, report the number of rows and columns.

Question 2**3 marks**

Show the data type of every column. Which columns are categorical?

Question 3**3 marks**

How many missing values are there in the whole dataset? How many rows are exact duplicates?

Question 4**4 marks**

For `price` and `carat`, report the mean, median, minimum and maximum. Is `price` right-skewed? Justify using the mean and the median.

Question 5**4 marks**

How many diamonds are there of each `cut`? Which cut is the most common, and what percentage of all diamonds is it?

Question 6**6 marks**

Compute the mean price for each `cut`, sorted from highest to lowest. The best cut (Ideal) has the lowest mean price. Compute the mean `carat` for each cut and use it to explain why.

Question 7**4 marks**

Plot a histogram of `price`. Describe its shape in one sentence.

Question 8**4 marks**

Draw a scatter plot of `carat` (x) against `price` (y). Use transparency or a random sample of 5,000 rows so the points are readable. Describe the relationship.

Question 9**4 marks**

Draw a box plot of `price` for each `color`. Which colour has the highest median price? (D is the best colour and J the worst.)

Question 10**6 marks**

Compute the correlation matrix of the numeric columns. Which variable is most strongly correlated with `price`, and what is the correlation? (Optional: show it as a heatmap.)

Dataset: `diamonds` on Kaggle (shivam2503/diamonds), the same data as seaborn's built-in copy: 53,940 diamonds with price, carat, cut, colour, clarity and dimensions. End of question paper.